

# HOSPITAL PATIENT RECORDS MANAGEMENT SYSTEM(HPRMS) USING MYSQL

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20/5/2026

# INTRODUCTION

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THIS PROJECT FOCUSES ON DESIGNING AND IMPLEMENTING A HOSPITAL PATIENT RECORDS MANAGEMENT SYSTEM USING **MYSQL**.

THE SYSTEM MANAGES INFORMATION ABOUT DOCTORS, PATIENTS, AND APPOINTMENTS WHILE SUPPORTING **CRUD** OPERATIONS AND ANALYTICAL REPORTING.

# PROBLEM STATEMENT

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HOSPITALS REQUIRE AN EFFICIENT SYSTEM FOR STORING, RETRIEVING ,  
UPDATING AND MANAGING PATIENTS AND APPOINTMENT RECORDS.  
CURRENT MANUAL METHODS MAY RESULT IN:

- MISSING PATIENT RECORDS.
- DUPLICATE ENTRIES.
- SLOW RETRIEVAL OF INFORMATION.
- POOR APPOINTMENT MANAGEMENT.

# OBJECTIVES

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- › TO DESIGN AND IMPLEMENT A HOSPITAL PATIENT RECORDS MANAGEMNET SYSTEM USING MYSQL.
  - › STORE PATIENTS' RECORDS.
  - › MANAGE DOCTOR INFORMATION.
  - › SCHEDULE APPOINTMENTS.
  - › PERFORM CRUD OPERATIONS.
  - › GENERATE REPORTS AND INSIGHTS.
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# SCOPE OF THE SYSTEM

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## › THE SYSTEM MANAGES:

- › PATIENT RECORDS
- › DOCTOR RECORDS
- › APPOINTMENT RECORDS

## › THE SYSTEM SUPPORTS:

- › ADDING NEW RECORDS
  - › VIEWING RECORDS
  - › SEARCHING RECORDS
  - › UPDATING RECORDS
  - › DELETING RECORDS
- 

# DATABASE DESIGN(ERD)

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## DOCTORS

DOCTOR\_ID ---- INT (PK)  
DOCTOR\_NAME ---  
VARCHAR(50)  
SPECIALIZATION--- VARCHAR(50)  
PHONE\_NUMBER ---  
VARCHAR(20)

## PATIENTS

PATIENT\_ID ---- INT (PK)  
PATIENT\_NAME ---  
VARCHAR(50)  
GENDER--- VARCHAR(10)  
AGE---INT  
DOCTOR\_ID---INT (FK)  
HOSPITAL\_BILL---DECIMAL(10,2)

## APPOINTMENT

APPOINTMENT\_ID --- INT(PK)  
PATIENT\_ID ---- INT(FK)  
DOCTOR\_ID---- INT(FK)  
APPOINTMENT\_DATE ---- DATE

# SQL IMPLEMENTATION/CRUD OPERATIONS

Limit to 1000 rows

1 • create database HOSPITAL\_DB;

2 • USE HOSPITAL\_DB;

3 • create table DOCTORS

```
(DOCTOR_ID int primary key,  
DOCTOR_NAME varchar(50),  
SPECIALIZATION varchar(50),  
PHONE_NUMBER varchar(20)  
);
```

10 • insert into doctors

```
values (101,"DR JOHN","CARDIOLOGY","08012345678"),  
(102,"DR MARY","PEDIATRICS","08023456789"),  
(103,"DR DAVID","NEUROLOGY","08034567890"),  
(104,"DR GRACE","ORTHOPEDICS","08045678901"),  
(105,"DR JAMES","DERMATOLOGY","08056789012");
```

60 • select \* FROM doctors;

| Result Grid | Filter Rows: | Edit:          | Exp          |
|-------------|--------------|----------------|--------------|
| DOCTOR_ID   | DOCTOR_NAME  | SPECIALIZATION | PHONE_NUMBER |
| 101         | DR JOHN      | CARDIOLOGY     | 08012345678  |
| 102         | DR MARY      | PEDIATRICS     | 08023456789  |
| 103         | DR DAVID     | NEUROLOGY      | 08034567890  |
| 104         | DR GRACE     | ORTHOPEDICS    | 08045678901  |
| 105         | DR JAMES     | DERMATOLOGY    | 08056789012  |

18 • create table PATIENTS

```
(PATIENT_ID int primary key,  
PATIENT_NAME varchar(50),  
GENDER varchar(10),  
AGE int,  
DOCTOR_ID int,  
foreign key (DOCTOR_ID)  
references doctors(DOCTOR_ID),  
HOSPITAL_BILL DECIMAL (10,2)  
);
```

29 • insert into patients

```
values(1,"JAMES","MALE",35,101,45000),  
(2,"SARAH","FEMALE",29,102,25000),  
(3,"MICHAEL","MALE",40,101,60000),  
(4,"BLESSING","FEMALE",22,103,30000),  
(5,"DANIEL","MALE",50,104,70000),  
(6,"MARY","FEMALE",31,105,35000),  
(7,"PETER","MALE",45,102,55000);
```

62 • select \* FROM patients;

|             |            |              |        |              |           |               |  |               |  |
|-------------|------------|--------------|--------|--------------|-----------|---------------|--|---------------|--|
| Result Grid |            |              |        | Filter Rows: |           | Edit:         |  | Export/Import |  |
|             | PATIENT_ID | PATIENT_NAME | GENDER | AGE          | DOCTOR_ID | HOSPITAL_BILL |  |               |  |
| ▶           | 1          | JAMES        | MALE   | 35           | 101       | 45000.00      |  |               |  |
|             | 2          | SARAH        | FEMALE | 29           | 102       | 25000.00      |  |               |  |
|             | 3          | MICHAEL      | MALE   | 40           | 101       | 60000.00      |  |               |  |
|             | 4          | BLESSING     | FEMALE | 22           | 103       | 30000.00      |  |               |  |
|             | 5          | DANIEL       | MALE   | 50           | 104       | 70000.00      |  |               |  |
|             | 6          | MARY         | FEMALE | 31           | 105       | 35000.00      |  |               |  |
|             | 7          | PETER        | MALE   | 45           | 102       | 55000.00      |  |               |  |

39 • create table APPOINTMENTS

```
(APPOINTMENT_ID INT primary key,  
PATIENT_ID int,  
foreign key (PATIENT_ID)  
references patients(PATIENT_ID),  
DOCTOR_ID int,  
foreign key (DOCTOR_ID)  
references doctors(DOCTOR_ID),  
APPOINTMENT_DATE date);
```

49 • insert into appointments

```
values(201,1,101,"2026-06-01"),  
(202,2,102,"2026-06-02"),  
(203,3,101,"2026-06-03"),  
(204,4,103,"2026-06-04"),  
(205,5,104,"2026-06-05"),  
(206,6,105,"2026-06-06"),  
(207,7,102,"2026-06-07");
```

64 • select \* FROM appointments;

| Result Grid    | Filter Rows: | Edit:     | Export           |
|----------------|--------------|-----------|------------------|
| APPOINTMENT_ID | PATIENT_ID   | DOCTOR_ID | APPOINTMENT_DATE |
| 201            | 1            | 101       | 2026-06-01       |
| 202            | 2            | 102       | 2026-06-02       |
| 203            | 3            | 101       | 2026-06-03       |
| 204            | 4            | 103       | 2026-06-04       |
| 205            | 5            | 104       | 2026-06-05       |
| 206            | 6            | 105       | 2026-06-06       |
| 207            | 7            | 102       | 2026-06-07       |

SIT TIGHT,,NEXT CHAPTER IN A BIT

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IN PART 2, I'LL SHOW THE SQL QUERIES AND  
BUSINESS INSIGHTS GENERATED FROM THIS  
DATABASE

**KeshDAnalyst**

# KEY METRICS

Limit to 1000 rows

```
128
129 • select
130   sum(HOSPITAL_BILL)
131   AS HOSPITAL_TOTAL_REVENUE
132   from patients;
```

Result Grid | Filter Rows: | Export: | Wrs

|   | HOSPITAL_TOTAL_REVENUE |
|---|------------------------|
| ▶ | 320000.00              |

```
74
75 • select avg(HOSPITAL_BILL)
76   from patients;
77
78
```

Result Grid | Filter Rows: |

|   | avg(HOSPITAL_BILL) |
|---|--------------------|
| ▶ | 45714.285714       |

```
70
71 • select count(*)
72   as TOTAL_NO_OF_PATIENTS
73   from patients;
74
```

Result Grid | Filter Rows: |

|   | TOTAL_NO_OF_PATIENTS |
|---|----------------------|
| ▶ | 7                    |

```
124
125 • select avg(AGE)
126   as AVERAGE_AGE
127   from patients;
128
```

Result Grid | Filter Rows: |

|   | AVERAGE_AGE |
|---|-------------|
| ▶ | 36.0000     |

```
128
129 • select count(*) APPOINTMENT_ID
130   FROM appointments;
131
132
```

Result Grid | Filter Rows: |

|   | APPOINTMENT_ID |
|---|----------------|
| ▶ | 7              |

```
125
126 • select count(*) DOCTOR_ID
127   from doctors;
128
129
```

Result Grid | Filter Rows: |

|   | DOCTOR_ID |
|---|-----------|
| ▶ | 5         |

# ANALYTICAL QUERIES

```
104 • select PATIENT_NAME,HOSPITAL_BILL
105 from patients
106 group by PATIENT_NAME,HOSPITAL_BILL
107 order by HOSPITAL_BILL asc
108 limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap

| PATIENT_NAME | HOSPITAL_BILL |
|--------------|---------------|
| SARAH        | 25000.00      |

```
109
110 • select DOCTOR_NAME,
111 count(*) patient_id
112 from doctors
113 join patients
114 ON doctors.DOCTOR_ID = patients.DOCTOR_ID
115 group by doctors.DOCTOR_NAME;
116
```

Result Grid | Filter Rows: | Export: | Wrap C

| DOCTOR_NAME | patient_id |
|-------------|------------|
| DR JOHN     | 2          |
| DR MARY     | 2          |
| DR DAVID    | 1          |
| DR GRACE    | 1          |
| DR JAMES    | 1          |

```
83 • select GENDER,
84 count(*) TOTAL_NO_OF_PATIENTS
85 from patients
86 group by GENDER;
87
```

| GENDER | TOTAL_NO_OF_PATIENTS |
|--------|----------------------|
| MALE   | 4                    |
| FEMALE | 3                    |

```
98 • select PATIENT_NAME,HOSPITAL_BILL
99 from patients
100 group by PATIENT_NAME,HOSPITAL_BILL
101 order by HOSPITAL_BILL desc
102 limit 1;
```

| PATIENT_NAME | HOSPITAL_BILL |
|--------------|---------------|
| DANIEL       | 70000.00      |

```
117 • select doctors.SPECIALIZATION,
118 count(patients.PATIENT_ID)
119 from doctors
120 inner join patients
121 on doctors.DOCTOR_ID = patients.DOCTOR_ID
122 group by doctors.SPECIALIZATION
123 order by count(patients.PATIENT_ID) desc
124 limit 1;
```

| SPECIALIZATION | count(patients.PATIENT_ID) |
|----------------|----------------------------|
| CARDIOLOGY     | 2                          |

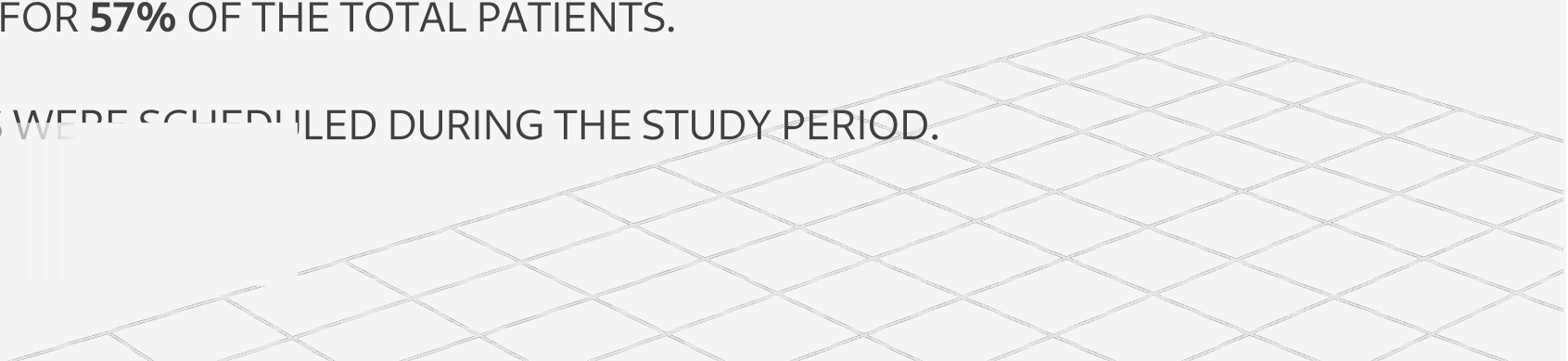
```
138 • select patients.PATIENT_NAME,
139 doctors.DOCTOR_NAME,
140 doctors.SPECIALIZATION,
141 appointments.APPOINTMENT_DATE
142 from patients
143 inner join doctors
144 ON patients.DOCTOR_ID = doctors.DOCTOR_ID
145 inner join appointments
146 ON doctors.DOCTOR_ID = appointments.DOCTOR_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| PATIENT_NAME | DOCTOR_NAME | SPECIALIZATION | APPOINTMENT_DATE |
|--------------|-------------|----------------|------------------|
| JAMES        | DR JOHN     | CARDIOLOGY     | 2026-06-01       |
| JAMES        | DR JOHN     | CARDIOLOGY     | 2026-06-03       |
| MICHAEL      | DR JOHN     | CARDIOLOGY     | 2026-06-01       |
| MICHAEL      | DR JOHN     | CARDIOLOGY     | 2026-06-03       |
| SARAH        | DR MARY     | PEDIATRICS     | 2026-06-02       |
| SARAH        | DR MARY     | PEDIATRICS     | 2026-06-07       |
| PETER        | DR MARY     | PEDIATRICS     | 2026-06-02       |
| PETER        | DR MARY     | PEDIATRICS     | 2026-06-07       |
| BLESSING     | DR DAVID    | NEUROLOGY      | 2026-06-04       |
| DANIEL       | DR GRACE    | ORTHOPEDICS    | 2026-06-05       |
| MARY         | DR JAMES    | DERMATOLOGY    | 2026-06-06       |

# KEY FINDINGS

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- THE TOTAL REVENUE GENERATED BY THE HOSPITAL IS **320,000**.
  - THE AVERAGE HOSPITAL BILL IS **45,700**.
  - **DANIEL** WAS THE HIGHEST PAYING PATIENT WITH **70,000** AND **SARAH** WAS THE LOWEST PAYING PATIENT WITH **25,000**.
  - **DR JOHN & DR MARY** MANAGED THE HIGHEST NUMBER OF PATIENTS.
  - **CARDIOLOGY** RECORDED THE HIGHEST PATIENT VOLUME.
  - MALE PATIENTS ACCOUNTED FOR **57%** OF THE TOTAL PATIENTS.
  - A TOTAL OF **7** APPOINTMENTS WERE SCHEDULED DURING THE STUDY PERIOD.
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# SKILLS DEMONSTRATED

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## TECHNICAL SKILLS

- MYSQL
- SQL QUERIES
- DATABASE DESIGN
- CRUD OPERATIONS
- JOINS
- AGGREGATED FUNCTIONS
- FOREIGN KEYS
- PRIMARY KEYS

## ANALYTICAL SKILLS

- DATA RETRIEVAL
- REPORTING
- DATA ANALYSIS

# CONCLUSION

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THE HOSPITAL PATIENT RECORDS MANAGEMENT SYSTEM SUCCESSFULLY DEMONSTRATES HOW RELATIONAL DATABASES CAN BE USED TO MANAGE HEALTHCARE RECORDS EFFICIENTLY. THROUGH THE IMPLEMENTATION OF TABLE RELATIONSHIPS, CRUD OPERATIONS, AND ANALYTICAL QUERIES, THE PROJECT HIGHLIGHTS THE PRACTICAL APPLICATION OF MYSQL IN REAL-WORLD HEALTHCARE DATA MANAGEMENT.

# RECOMMENDATIONS

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- IMPLEMENT USER AUTHENTICATION AND AUTHORIZATION.
- INTEGRATE BILLING AND PAYMENT MODULES.
- ADD ELECTRONIC MEDICAL RECORD (EMR) FUNCTIONALITY.
- DEVELOP A DASHBOARD USING EXCEL OR POWER BI FOR VISUALIZATION.
- IMPLEMENT AUTOMATED APPOINTMENT REMINDERS.